

Common Trend VAR with $I(1)$ and $I(2)$ series

December 8, 2024

Abstract

This paper presents a common trend VAR model, which allows us to estimate trend and cyclical components of output, consumption, inflation, and interest rates. From a theoretical perspective, we build upon the existence of a balanced growth (permanent income hypothesis) and the consumption Euler equation. The permanent income hypothesis and balanced growth implies cointegration between (log) consumption and (log) output. Furthermore, the consumption Euler equation implies an existence of a cointegration relation between the consumption growth rate and the real interest rate. This approach also allows us to compute the natural/equilibrium rate of interest, which is a central concept in macroeconomics. The analysis is performed on the US data ranging from 1960 (Q1) to 2022 (Q3).

C32, C82, E32

Common Trend VAR, Equilibrium Interest Rate, Consumption Euler Equation.

1 Introduction

The study presents a common trend VAR model in the spirit of Del Negro *et al.* (2017, 2019) to study the relationship between output, consumption, inflation, and interest rates. The methodology explicitly reflects slow-moving trends in individual variables, significantly improving the VAR model interpretability due to the decomposition into trend and cyclical components.

More specifically, the explicit treatment of trend components allows us to study the implications of the permanent income hypothesis and a balanced growth path. By acknowledging the possibility of fewer trends than variables, the model accommodates steady-state relationships among these variables. This is achieved by imposing the restriction that (log) consumption is 'nearly' cointegrated with (log) output. Essentially, this approach facilitates the exploration of the time series' characteristics and acknowledges that trend elements encompass stochastic factors, aligning with the concept of persistent shocks within an economy (Stock & Watson, 1988; Warne, 1993). Moreover, the proposed framework integrates additional constraints deduced from the consumption Euler equation. This subsequently allows for the computation of the natural/equilibrium interest rate, a central concept in macroeconomics. We define equilibrium real interest rate as a long-run notion that aligns with the equilibrium between savings and investment over periods beyond the business-cycle frequency as in Kiley (2020a).

From an econometric perspective, the common trend VAR is a multivariate trend-cycle decomposition that represents an alternative to other commonly used univariate or multivariate filters. For example, studies using multivariate trend-cycle decomposition are various unobserved components and 'semi-structural' models used by central banks and other practitioners, for example, Beneš *et al.* (2010), Tóth (2021). Other prominent and closely related examples of trend-cycle decomposition are studies based on vector error correction (VECM) and the multivariate Beveridge Nelson decomposition (Engle & Granger (1987), Garratt *et al.* (2006), Murasawa (2015), Morley & Wong (2020), Morley *et al.* (2023), and many others), as any non-stationary series can be decomposed into trend and cyclical components.¹ The most similar frameworks to the common trend VAR are trend-cycle VAR by Andrle *et al.* (2013) and steady-state VAR by Villani (2009). The main difference between our approach and the ones above is that steady-states are not explicitly defined but are estimated and that we allow for them to be time-varying. In this regard, we are also generally inspired by the shifting endpoint literature, exemplified by studies conducted by Kozicki & Tinsley (2001), Cieslak & Povala (2015), Bauer & Rudebusch (2020). These studies generally show that empirical properties (interpretability and predictions) improve significantly after including shifting endpoints, i.e., detrending. In more detail, Kozicki & Tinsley (2001) demonstrate that the inflation trend can explain a substantial portion of inflation rate variation. The study by Cieslak & Povala (2015) presents similar findings for bond yields. Bauer & Rudebusch (2020) argue for including the change in real interest rate trend to fully capture the trend component in interest rates, as accounting solely for the inflation trend leaves a highly persistent component of interest rates unexplained.

¹For this reason the model can be parameterized using VAR, VECM, Beveridge Nelson decomposition or common trend representation.

2 Motivation

The main advantage of our approach is the explicit treatment of slow-moving trends in individual variables. This allows us to apriori account for possible cointegration relations based on economic theory. We motivate our model specification by permanent income hypothesis and consumption Euler equation or, more generally, by the existence of balance growth and the stability of great ratios.² However, since our study also includes nominal variables (inflation and nominal interest rate), it can be stated that the model specification is loosely inspired by the standard new Keynesian, for example, Smets & Wouters (2007).

To elaborate, we primarily build upon the permanent income hypothesis, which in the simplest form predicts that consumption is (close to) a random walk and that consumption and total income are cointegrated, and, therefore, the consumption output ratio is stable over time.³ To be precise, the permanent income hypothesis does not predict the exact cointegration relation of consumption and income (Deaton, 1992). However, with some approximations, we can consider this a permanent income-like relation between consumption and total income, where consumption-income cointegration is a limiting case. Following Campbell & Mankiw (1989) and Cochrane (1994) among others, we see this hypothesis as an approximation to log consumption and log output development.

A significant empirical caveat is the fact that there exists nearly perfect equality between income and consumption growth rates (Carroll & Summers, 1991). This finding goes against the standard implication of the traditional interpretation of the permanent income hypothesis that consumption growth and anticipated income growth are unrelated.⁴ Carroll *et al.* (1992); Carroll (2022) show that this behaviour can be explained by the buffer-stock savings model, which they consider an alternative/more accurate description of the permanent income hypothesis. Nevertheless, using this approach, if gross wealth is at the target ratio, then the expected consumption growth rate is approximately equal to the growth rate of permanent income (and convexity adjustment), which suggests the emphasis should be placed on growth rate relations.

To reconcile the different perspectives, we model (log) consumption and (log) output using local linear trend trends.⁵ The local linear trend model assumes that both the mean and the slope of the trend follow random walks. However, both (log) consumption and (log) output specification share the same trend growth rate (slope). This generally allows for more flexibility as the consumption-output ratio can slowly change over time. Depending on the ratio of variances of the error terms, this process will either be closer to a cointegration relation or two independent random walks. By imposing strong priors, we ensure that the consumption-output ratio remains relatively stable, effectively constraining log consumption to be 'nearly' cointe-

²These explanations are not mutually exclusive.

³Cochrane (1994) emphasize that consumption is nearly a random walk, and if consumption does not change, agents must consider any fluctuation in output as transitory. Therefore, by observing consumption, it is possible to separate output into 'perceived' permanent and cyclical components.

⁴An associated point to note is that the standard interpretation of the permanent income hypothesis can account for the smoothness of consumption only if changes in income are largely predictable.

⁵The implicit assumption is that (log) consumption and (log) output are I(2) processes.

grated with log output. The approach also allows us to partially avoid arbitrary smoothness trend estimates as it allows for more flexibility. This is similar to allowing for structural breaks, the correlation between trend and cyclical components, or allowing for state switching as in Kim *et al.* (2007), Morley (2007), Attfield & Temple (2010). Also, for example, Whelan (2003) stresses the importance of studying great ratios in terms of current prices, as nominal shares will be stationary under more general conditions, which is another reason for allowing more flexibility.

An alternative motivation for our model arises from the concept of a balanced growth path.⁶ The balanced growth path is often explained by assuming the existence of a common stochastic productivity trend in a (closed single-sector) economy. To be more specific, similar to King *et al.* (1991), we can assume the existence of an exogenous stochastic process for total factor productivity⁷ and given these assumptions, (log) consumption is cointegrated with (log) output, with a unit cointegrating vector. As explained above, to allow for some flexibility, we assume the logarithm of output and consumption can be described by local linear trend models with an average (time-varying) growth rate of the total factor common to both processes.

Our model specification's second main theoretical foundation is the consumption Euler equation, which states a cointegration relationship between the real consumption growth rate and the real interest rate (Laubach & Williams, 2003; Holston *et al.*, 2017). To be precise, this is again not an exact relation but a limiting case (approximation) of a 'true' structural relation. Euler equation can be written as follows:

$$r_{t,equilibrium} = \delta + \rho g_{t,\lambda} \quad (1)$$

where $r_{t,equilibrium}$ denotes the natural/equilibrium real interest rate of interest, $g_{t,\lambda}$ is the expected trend growth rate of consumption, which in our specification also coincide with time-varying total factor productivity growth rate. δ denotes the discount rate, and ρ is the measure of relative risk aversion, and it is estimated to be a very close one as is shown in meta-study by Elminejad *et al.* (2022).⁸ This suggests a one-to-one relation between GDP growth and real interest rate. However, in this specification, we abstract from demographic changes and labour force development, which, according to many studies, are suggested to be one of the main drivers of the decreasing equilibrium interest rate, for example, Rachel & Smith (2017) or Cesa-Bianchi *et al.* (2022). Another possible modification, from which we abstract, is to introduce liquidity constraints, habit formation, sticky information or trend changes in wealth or markup evolution, etc. (Hall, 1978; Campbell & Cochrane, 1999; Sommer, 2007; Mian *et al.*, 2021; Ball & Mankiw, 2023). Most importantly, given the fact that consumption growth is uncertain, we should also introduce the convexity adjustment term (precautionary savings motive) (Carroll *et al.*, 1992;

⁶Along a balanced growth path, theory predicts that the equilibrium great ratios should be stable functions of structural parameters, and if these parameters are themselves stable, we expect the great ratios to be stationary. For theoretical treatment, see King *et al.* (1988, 2002).

⁷Authors presume log-utility, full-depreciation of the capital stock, and that output is produced via a constant returns-to-scale Cobb-Douglas production function and that total factor productivity follows the logarithmic random walk. The results can, however, be generalized under a quadratic approximation of the value function.

⁸The study updates earlier results by Havranek *et al.* (2015).

Carroll, 2022). These other factors are, however, implicitly included in our empirical estimates of δ . For this reason, we denote the empirical estimate of δ as $z_{t,trend}$, and we name it as an additional trend regarding real interest rate evolution and allow it to vary in time.

If we assume that the long-run consumption growth (total factor productivity growth rate) follows the I(1) process⁹. In this case, the Euler equation implies that the I(1) process also describe real interest rate development. In other words, the Euler equation means that the long-run expected consumption growth and the real interest rate must be of the same order of integration.

In our model specification, we further assume that inflation and nominal interest rate can be described as I(1) series as in Kozicki & Tinsley (2001).¹⁰ In more detail, we postulate that inflation, inflation expectations, and nominal interest rate load upon a long-run inflation trend, representing a steady-state expectation of inflation. A possible criticism of our approach is that the FED following inflation target suggests that inflation is stationary. However, this was not always the case, and the inflation targeting itself depends on the FED's credibility. Therefore, as the Fisher and Euler equations suggest, the nominal interest rate loads upon both the inflation and consumption growth rate trends. As discussed above, we also assume that interest rate loads upon an additional trend regarding (real) interest rate evolution ($z_{t,trend}$). This factor accounts for other factors not considered in the Euler or Fischer equation. From a practical point of view, including this additional trend means that low-frequency fluctuations not explained by output growth or inflation will likely be explained by this trend (Buncic, 2021).

3 Data and Methodology

The article uses data from 1947 (Q1) to 2022 (Q3)¹¹ regarding output, consumption, interest rate, inflation, and inflation expectations. Specifically, we use the following data: real GDP and real consumption in levels, three-month Treasury Bill rate, Personal consumption expenditures price index (PCE), and median value regarding long-run PCE from the Survey of Professional Forecasters (SPF) as a measure of inflation expectation. All data were obtained from FRED¹² except for inflation expectations, which are obtained from the SPF. Before the year 2007 long-run, we take inflation expectation from Del Negro *et al.* (2017) as the series is unavailable on the SPF. This series is the survey-based long-run (5 to 10 years ahead) PCE inflation expectations series of the Federal Reserve Board's FRB/US econometric model.

We use seasonally adjusted end-of-month values of quarterly data (with the exception of the three-month Treasury Bill rate and inflation expectations, which are not seasonally adjusted). We multiply the log differences of the Personal consumption expenditures price index (PCE) by 400 to get an annualized rate, and we also take a log transformation of output and consumption

⁹See Cerra *et al.* (2023), where authors provide a comprehensive summary of discussion regarding output being trend stationary or the difference stationary.

¹⁰This is generally confirmed by ADF and KPSS test results as is shown later.

¹¹The period from 1947 (Q1) to 1959 (Q4) is used as a training period. The analysis is performed on the US data ranging from 1960 (Q1) to 2022 (Q3).

¹²Following tickers from FRED: GDPC1, PCECC96, DPCERD3Q086SBEA, TB3MS.

and multiply them by 100. To simplify notation, let's introduce $X_t = (y_t, c_t, \pi_t, \pi_t^e, i_t)$, where y_t denote log real output, c_t is log real consumption, π_t is inflation, π_t^e represent expected inflation, and i_t denote nominal interest rate.

Additional information regarding the data used can be found in Table 1. The table shows the basic summary statistics regarding the final data entering the estimation, where we compute the mean, standard deviation, min and max values, and autocorrelation coefficient at first lag. We also display Dickey-Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test results. As expected, the standard unit root tests mainly indicate the time series' nonstationarity in levels. We further show summary statistics of various transformations of the variables used, including first and second differences of the series and consumption-output ratio. The results generally support our modeling choice of consumption and output, being I(2) processes combined with relatively strong priors that these series are 'nearly' cointegrated.

Table 1: Summary Statistics - Variables

Variable	Mean	Sd	Max	Min	Rho	ADF	KPSS	N
y	912.37	53.08	991.33	808.15	0.99	0.60	0.01	252
c	869.33	57.24	956.47	760.03	0.99	0.61	0.01	252
i	4.33	3.17	14.34	-0.01	0.96	0.06	0.01	252
π	3.25	2.57	12.01	-6.38	0.82	0.10	0.01	252
π^e	3.07	1.53	7.72	1.68	0.99	0.62	0.01	252
Δy	0.72	1.10	7.56	-8.87	0.02	0.01	0.05	251
Δc	0.78	1.08	8.94	-9.67	-0.08	0.01	0.04	251
Δi	0.01	0.89	3.56	-6.36	-0.12	0.01	0.10	251
$\Delta \pi$	0.01	1.55	5.14	-10.65	-0.26	0.01	0.10	251
$\Delta \pi^e$	0.00	0.16	0.83	-0.57	0.16	0.01	0.10	251
$\Delta^2 y$	0.01	1.54	16.43	-7.68	-0.55	0.01	0.10	250
$\Delta^2 c$	-0.00	1.59	18.61	-8.08	-0.58	0.01	0.10	250
$c - y$	-43.04	4.36	-34.43	-51.37	0.98	0.04	0.01	252

Note: Rho denotes the autocorrelation coefficient at first lag. N denotes the number of observations. ADF and KPSS denote p-values of Augmented Dickey-Fuller and Kwiatkowski–Phillips–Schmidt–Shin statistics. The number of lags in ADF and KPSS tests was selected to minimize the AIC. The analysis is performed on the data ranging from 1960 (Q1) - 2022 (Q3).

4 Methodology

Given the assumed linearity of our model specification, the model can be easily cast in the state space form. The measurement equation can be written as follows:

$$X_t = \Lambda X_{t,trend} + X_{t,cycle} \quad (2)$$

where Λ denotes the matrix of loadings, which determines the number of common trends. $X_{t,i}$ denote the states, where the cyclical component is denoted by a subscript 'cycle', the trend component by a subscript 'trend', and both $X_{t,trend}$ and $X_{t,cycle}$ are latent. Also, note that we do not allow error terms in measurement equations. The equation 2 can be rewritten in detail as follows:

$$y_t = \lambda_{t,y,trend} + y_{t,cycle} \quad (3)$$

$$c_t = \lambda_{t,c,trend} + c_{t,cycle} \quad (4)$$

$$i_t = \pi_{t,trend} + z_{t,trend} + 4g_{t,\lambda} + i_{t,cycle} \quad (5)$$

$$\pi_t^e = \pi_{t,trend} + \pi_{t,cycle}^e \quad (6)$$

$$\pi_t = \pi_{t,trend} + \pi_{t,cycle} \quad (7)$$

where $z_{t,trend}$ is the additional trend regarding real interest rate evolution and $g_{t,\lambda}$ denotes the long-run expected (time-varying) growth rate of output and consumption, which can be interpreted as expected total factor productivity growth rate. This trend growth rate is multiplied by 4, given the fact that we use the quarterly growth rate, and inflation and interest rates are annualized. The following equations then describe the trend component's evolution:

$$z_{t,trend} = z_{t-1,trend} + e_{t,r} \quad (8)$$

$$\pi_{t,trend} = \pi_{t-1,trend} + e_{t,\pi} \quad (9)$$

$$\lambda_{t,y,trend} = \lambda_{t-1,y,trend} + g_{t,\lambda} + e_{t,y} \quad (10)$$

$$\lambda_{t,c,trend} = \lambda_{t-1,c,trend} + g_{t,\lambda} + e_{t,c} \quad (11)$$

$$g_{t,\lambda} = g_{t-1,\lambda} + e_{t,g} \quad (12)$$

where $e_{t,i}$ denote i.i.d. error terms regarding trend evolution. In summary, all observed series load upon at least one trend and one cyclical component. Specifically, we assume the existence of a stochastic trend regarding inflation, stochastic trends regarding consumption and output with the same slope, and an additional/residual stochastic trend regarding real interest rate evolution.

The model is estimated using Bayesian methods, and the algorithm is taken directly from Del Negro *et al.* (2017). In more detail, the model is estimated simulating 50000 draws with a burn-in of 25000. The initial conditions regarding trend components are based on pre-sample averages and univariate detrending. The prior regarding the variance-covariance matrix of the trend innovation is very conservative. The covariance matrix is diagonal with the following elements: (2/400, 0.5/400, 100/400, 100/400, 1/400). We use $\kappa = 100$ as in Del Negro *et al.* (2017). This prior on the covariance matrix implies that the standard deviation of the expected change in trend inflation over one century is approximately two percentage points and one percentage point regarding the additional interest rate trend. The standard deviation of the quarterly output and consumption trend growth rate is one-half of a percentage point in a century, i.e. approximately two percentage points regarding the annual growth rate. The standard deviation regarding the mean of the local linear trend models regarding consumption and output is, in both cases, 100/400. To put it plainly, we use relatively strong priors regarding the variability of these trends to ensure that they 'do not move much'.

The cycle components $X_{t,cycle}$ are modeled using a standard unrestricted Bayesian VAR(2) model, where we use two lags and standard Minnesota-style priors specified as a normal inverse Wishart prior. These are centered on zero (coefficient of the first own lag) because the variables are in gap form. We use the standard values of all the hyperparameters. Thus, the overall tightness is set to 0.2, and the scaling coefficient controlling the speed of lag decay is set to 2

and prior for the covariance matrix is a relatively uninformative inverse Wishart distribution centered at a diagonal matrix with the following elements: (2/400, 1.5/400, 100/400, 100/400, 1.5/400). The BVAR can be written as follows:

$$X_{t,cycle} = C + A_1 X_{t-1,cycle} + A_2 X_{t-2,cycle} + \epsilon_t \quad (13)$$

A_1, A_2 are matrices of coefficients, C is constant matrix, and ϵ_t denotes of error terms. We also do not allow for any correlation between ϵ_t and $e_{t,i}$. This relatively strong restriction is criticized by Morley (2007) as economic theory does not rule out the presence of a correlation between permanent and transitory shocks, i.e. 'time to-build' effects. However, the use of local linear trends allows us to at least partially mitigate arbitrary smoothness trend estimates. This is because the use of local linear trends has a qualitatively similar effect to specifying a more flexible trend process. We also stress that all cyclical components are treated as endogenous in the model.

The intuition behind our approach is that the long-horizon forecast only reflects its trend value as the cyclical component is assumed to be zero. Therefore, for large n , it can then be written that the best n -period ahead forecast given by the trend component, i.e., $\lim_{n \rightarrow \infty} X_{t+n} = X_{t+n,trend}$. Regarding inflation and interest rate, the optimal long-run forecast is the last known trend value since we assume that they follow the I(1) process. In the case of output and consumption, the long-run forecast is the following: $y_{t+n} = ng_{t,\lambda} + \lambda_{t,y,trend}$, and $c_{t+n} = ng_{t,\lambda} + \lambda_{t,c,trend}$.

Given the model specification, we can also easily determine the real and the nominal equilibrium interest rate ($r_{t,equilibrium}$, $i_{t,equilibrium}$):

$$r_{t,equilibrium} = z_{t,trend} + 4g_{t,\lambda} \quad (14)$$

$$i_{t,equilibrium} = \pi_{t,trend} + z_{t,trend} + 4g_{t,\lambda} \quad (15)$$

Our specification is from methodological point of view essentially the same as in Laubach & Williams (2003), which enables a direct comparison with their results.

5 Results

Figure 2 displays the estimated trend components and equilibrium interest rates.¹³ Regarding trend evolution, trend inflation surged during the 1970s energy crises but declined after the 1981-82 recession, stabilizing around two percent in the past two decades due to inflation targeting. As for the trend growth rate of output and consumption (total factor productivity growth rate), we can observe a gradual slowdown since the 1980s. However, the total factor productivity growth rate displays relatively large variability and significantly fluctuates over time, which suggests a large number of innovations/shocks in this estimated permanent component. In other words,

¹³Given the fact that these are model-dependent results as both permanent and transitory components are not directly observed, we also show 68% credible intervals.

the trend component is relatively volatile and is constantly being changed by new shocks. This is generally in line with Real Business Cycle (RBC) reasoning, which claims that exogenous (technology) shocks play a central role in economic fluctuations in industrial economies.

We further document a continuous decline in both equilibrium nominal and real interest rates since the early 1980s, aligning with general intuition. This decline in the equilibrium nominal interest rate¹⁴ is attributed to descending trends in inflation and the trend growth rate of output and consumption (total factor productivity growth rate), as well as additional long-term trends regarding real interest rate evolution. Notably, the decline in equilibrium real interest rates, particularly in the past two decades characterized by predominantly negative real equilibrium interest rates, cannot be fully explained solely by the deceleration in the trend growth rate of output and consumption or a reduction in long-term inflation expectations. Instead, it reflects a substantial decrease in additional long-term trends regarding real interest rate evolution. Furthermore, our findings also reveal that the evolution of the real equilibrium interest rate is not solely characterized by a relatively linear descent, but the equilibrium rate significantly fluctuates over time, mainly due to the variability in the trend growth rate of output and consumption. However, the additional trend component in the real interest rate also exhibits notable movements, indicating a relatively large increase in the real equilibrium interest rate from the 1960s to the 1980s.

Comparing our model estimates to that produced by the Laubach & Williams (2003) can provide further insights into our results. Notably, our results regarding equilibrium interest rate evolution significantly differ from Laubach & Williams (2003). The difference is mainly caused by the different behavior of the conceptually same estimate regarding the additional long-term trend regarding real interest rate evolution. This can be explained by the fact that all variables are endogenous in the model, and our trend specification is generally less restrictive.

The posterior estimate of the covariance matrix also suggests a significant correlation between trends despite the fact that we a priori assume that the trends are independent.¹⁵ More specifically, a relatively strong correlation exists between the additional trend regarding interest rate evolution and both trend inflation and output growth trends (in both cases, the correlation is approximately 0.5). On the contrary, the correlation between these two trends (inflation and growth rate of output and consumption growth rate) is relatively small (approximately 0.2). These results indicate that real and nominal variables are generally independent in the long run. However, the results also suggest that the real equilibrium interest rate generally reacts more than one-to-one to changes in trend growth rate. The possible explanation is that change in discount rate δ is correlated with the trend growth rate of output and consumption movements or the economic agents are generally more risk averse, i.e., ρ is greater than 1.¹⁶ Our preferred explanation is that the additional trend regarding interest rate evolution represents some fac-

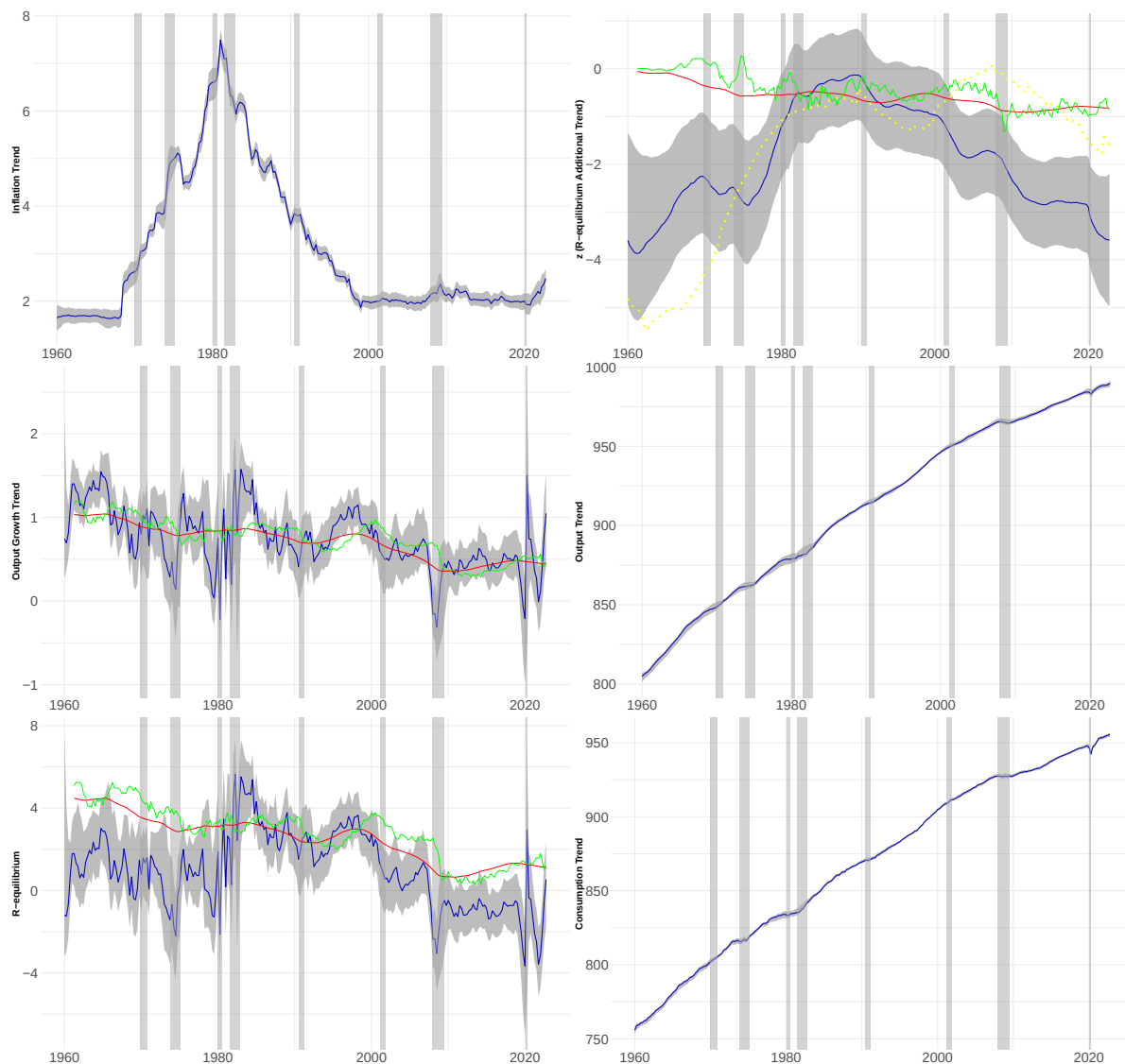
¹⁴The equilibrium nominal interest rate evolution is dominated by trend inflation development, which shows somewhat greater variability than the trends describing equilibrium interest rate evolution.

¹⁵The correlation reflects some form of unmodeled causality, as in a fully specified economic model, the underlying structural shocks should be uncorrelated. However, given that we are interested in trend behaviors, we can remain agnostic about the source of a correlation.

¹⁶In our estimation, we impose ρ equal to 1.

tor that was omitted in our specification of the Euler equation, which is correlated with the output/consumption trend growth rate. The development of the additional interest rate trend indicates a possibility that it can be partly explained by demographic changes (yellow dots), which exhibit a broadly similar trend.¹⁷

Figure 1: Trend components



Note: The gray bands denote 68% credible intervals. Blue line - median. Red line - 'one-sided' Laubach & Williams (2003) estimate. Green line - 'two-sided' Laubach & Williams (2003) estimate. Yellow dots - Ratio of US working age population (Aged 15-64) to total US population (secondary axis). The analysis is performed on the data ranging from 1960 (Q1) - 2022 (Q3).

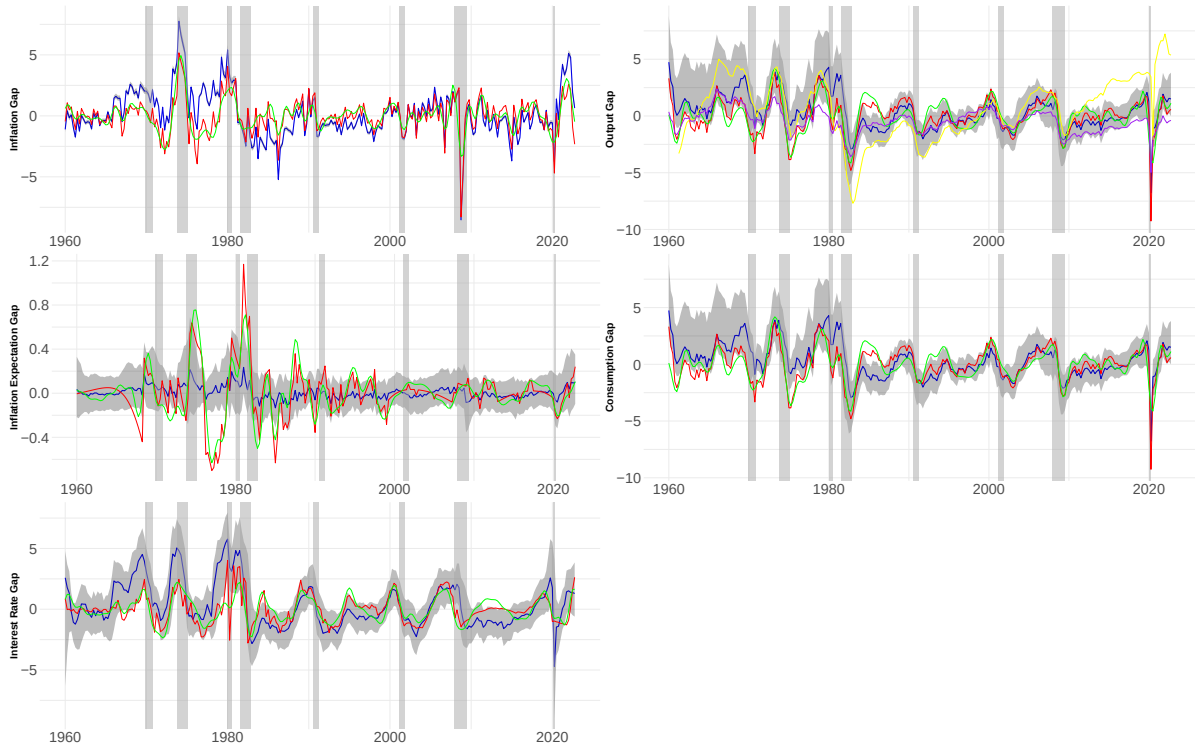
Figure 2 displays the estimated cyclical/gap components. To get intuition regarding cyclical component estimates, we compare our cyclical estimates to the results of other usually used detrending methods. To extract fluctuations regarding business cycle fluctuations, we use the

¹⁷'Baby boom generation' passing through the age distribution affects asset prices and equilibrium rates of return. See Mian *et al.* (2021).

Christiano-Fitzgerald filter (Christiano & Fitzgerald, 2003), which is a band-pass filter, where we focus on periods between 6-32 quarters. As a robustness check, we use the Hodrick-Prescott filter (Hodrick & Prescott, 1997) with lambda equal to 1600. Regarding the output gap, we also use an estimate from the Congressional Budget Office (CBO) as in Kiley (2020b).

The main finding is that our cyclical estimates obtained from common trend VAR generally correspond to other gap estimates obtained from the commonly used univariate filters. Most importantly, this result also applies to the consumption and output gaps, which were obtained assuming them being $I(2)$ series. This is in contrast to the Laubach & Williams (2003) output gap estimate (yellow line), as their calculation significantly differs from other commonly used methods. One exception is the behavior of the cyclical component of inflation expectations, which in our model is close to white noise and significantly differs from results obtained from univariate filters. However, extracting cyclical components regarding long-term expectations is a complicated task, and the obtained behavior can be easily defended as there is no general academic consensus regarding the cyclical properties of long-term expectations. The results also have a relatively important practical meaning in the sense that if we want to focus primarily on the development of the cyclical component. Univariate results give a fairly similar picture with significantly less computational burden.

Figure 2: Cyclical components



Note: The gray bands denote 68% credible intervals. Blue line - median. Red line - Hodrick-Prescott filter. Green line - Christiano-Fitzgerald filter. Yellow line - 'two-sided' Laubach & Williams (2003) estimate. Purple line - Congressional Budget Office estimate. The analysis is performed on the data ranging from 1960 (Q1) - 2022 (Q3).

Table 2 displays additional information regarding the cycle and trend estimates described

above and allows for a more formal analysis. The table shows the basic summary statistics regarding the estimated trend and cyclical components. We again compute the mean, standard deviation, min and max values, and autocorrelation coefficient at first lag.

Table 2: Summary Statistics - Estimated Components

Variable	Mean	Sd	Max	Min	Rho
$\pi_{t,trend}$	3.07 (2.89,3.25)	1.52 (1.48,1.55)	7.53 (7.19,7.79)	1.65 (1.46,1.80)	0.99 (0.99,0.99)
$z_{t,trend}$	-1.56 (-2.48,-0.64)	0.88 (0.94, 0.87)	-0.14 (-0.95,0.73)	-3.36 (-4.71,-2.16)	0.99 (0.99,0.99)
$\lambda_{t,y,trend}$	912.32 (910.49,914.09)	53.21 (53.68,52.95)	989.33 (986.41,991.15)	812.52 (809.08,814.9)	0.99 (0.99,0.99)
$\lambda_{t,c,trend}$	869.30 (867.78,870.73)	57.33 (57.72,57.13)	955.24 (952.80,956.48)	761.68 (759.37,763.73)	0.99 (0.99,0.99)
$g_{t,\lambda}$	0.70 (0.38,1.02)	0.31 (0.33,0.34)	1.79 (1.20,2.48)	-0.06 (-0.88,0.28)	0.75 (0.80,0.68)
$\Delta\lambda_{t,y,trend}$	0.70 (0.71,0.70)	0.36 (0.42,0.34)	1.97 (3.04,1.68)	-0.88 (-2.13,-0.10)	0.63 (0.42,0.73)
$\Delta\lambda_{t,c,trend}$	0.77 (0.77,0.77)	0.68 (0.75,0.62)	4.85 (6.19,3.47)	-4.55 (-5.84,-3.10)	0.04 (-0.08,0.15)
$\pi_{t,cycle}$	0.18 (0.00,0.36)	1.94 (1.91,1.97)	7.72 (7.30,8.12)	-8.55 (-8.74,-8.36)	0.69 (0.68,0.70)
$c_{t,cycle}^e$	0.00 (-0.18,.)	0.04 (0.06,0.10)	0.20 (-0.06,0.59)	-0.11 (-0.40,0.06)	0.58 (0.71,0.87)
$i_{t,cycle}$	0.05 (-1.39,1.44)	1.44 (1.40,1.82)	4.53 (1.37,7.31)	-3.72 (-8.42,-0.91)	0.92 (0.84,0.93)
$y_{t,cycle}$	0.06 (-1.71,1.89)	2.14 (2.29,2.65)	6.15 (2.80,10.40)	-6.55 (-9.26,-4.02)	0.90 (0.90,0.94)
$c_{t,cycle}$	0.04 (-1.40,1.55)	1.39 (1.27,2.02)	3.99 (1.36,8.34)	-4.52 (-7.30,-2.10)	0.91 (0.90,0.96)

Note: Rho denotes the autocorrelation coefficient at first lag. N denotes the number of observations. ADF and KPSS denote p-values of Augmented Dickey-Fuller and Kwiatkowski-Phillips-Schmidt-Shin statistics. The number of lags in ADF and KPSS tests was selected to minimize the AIC. The brackets shows 68% credible intervals computed from 25000 draws. The analysis is performed on the data ranging from 1960 (Q1) - 2022 (Q3).

The common trend VAR can also be used for forecasting; thus, we can also easily compute the forecast error variance decomposition (FEVD). This can be considered an intuitive way to measure the relative importance of trend (permanent) and cyclical (transitory) shocks. We only focus on transitory and permanent shocks in total, as the covariance (impact) matrix is non-recursive. However, transitory and permanent shocks are independent due to our modeling choice. Therefore, we can still obtain meaningful forecast error decomposition even without additional assumptions.

The FEVD results are displayed in Table 3. Our results show that the contribution of permanent and transitory innovations differs significantly for the variables considered. Inflation is generally defined by transitory shocks, and the contribution of transitory shocks is relatively high even with increasing forecast horizon. Regarding the output, the contributions are relatively balanced in the first forecast period. After that, the importance of cyclical shocks is diminishing relatively quickly. On the contrary, inflation expectations, consumption, and interest rates are forecast errors explained primarily by permanent shocks, even regarding short-term forecast horizons. The important finding is that innovations in the trend component explain most of the variability in interest rates. This, in combination with the clearly defined cyclical behavior of interest rates displayed above, suggests that the behavior of the cyclical part of interest rates is highly predictable.

The result obtained can be partially clarified by the permanent income hypothesis. Similarly to Cochrane (1994), we show that output behavior is characterized by a large share of transitory shocks. Regarding one-step-ahead prediction errors the contributions of cyclical and permanent

shocks are divided approximately equally. In other words, shocks to output holding other variables constant contain a significant transitory component, which accounts for large fractions of the variance of output growth. On the other hand, consumption behavior can be described as close to a ('nearly') random walk, as most of the forecast error variations are due to permanent shocks. Specifically, approximately ninety percent of the variance of one-step-ahead prediction errors regarding consumption is due to permanent shocks, which sharply contrasts with output results.

Table 3: Forecast Error Variance Decomposition

Horizon	π		π^e		i		y		c	
	Trend	Cycle	Trend	Cycle	Trend	Cycle	Trend	Cycle	Trend	Cycle
1	3.2 (2.2,5.9)	96.8 (94.1,97.8)	83.8 (76.0,90.6)	16.2 (9.4,24.0)	99.3 (98.2,99.8)	0.7 (0.2,1.8)	49.6 (30.7,68.7)	50.4 (31.3,69.3)	91.3 (84.5,96.0)	8.7 (4.0,15.5)
2	5.1 (3.4,8.7)	94.9 (91.3,96.6)	91.0 (79.5,93.6)	9.0 (6.4,20.5)	99.5 (98.6,99.9)	0.5 (0.1,1.4)	85.4 (77.4,92.2)	14.6 (7.8,22.6)	95.9 (91.2,98.3)	4.1 (1.7,8.8)
3	6.8 (4.5,11.1)	93.2 (88.9,95.5)	93.7 (81.5,95.0)	6.3 (5.0,18.5)	99.6 (98.8,99.9)	0.4 (0.1,1.2)	95.4 (91.4,97.6)	4.6 (2.4,8.6)	98.2 (95.5, 99.3)	1.8 (0.7,4.5)
4	8.5 (5.5,13.4)	91.5 (86.6,94.5)	95.2 (83.4 (88.3,97.7)	4.8 (4.0,16.6)	99.7 (99.0,99.9)	0.3 (0.1,1.0)	98.0 (95.7,4.3)	2.0 (1.1,4.3)	99.1 (97.5,99.6)	0.9 (0.4,2.5)
8	15.0 (9.2,21.3)	85.0 (78.7,90.8)	97.5 (88.3,97.7)	2.5 (2.3,11.7)	99.8 (99.4,99.9)	0.2 (0.1,0.6)	99.8 (99.2,99.9)	0.2 (0.1,0.8)	99.9 (99.5,99.9)	0.1 (0.1,0.5)
12	20.7 (12.4,28.2)	79.3 (71.8,87.6)	98.3 (91.0,98.4)	1.7 (1.6,9.0)	99.9 (99.5,99.9)	0.1 (0.1,0.5)	99.9 (99.7,100)	0.1 (0.0,0.3)	100 (99.8,100)	0 (0,0.2)
16	25.8 (15.3,34.1)	74.2 (65.9,84.7)	98.7 (92.7,98.8)	1.3 (1.2,7.3)	99.9 (99.6,100)	0.1 (0,0.4)	100 (99.9,100.0)	0 (0,0.1)	100 (99.9,100)	0 (0,0.1)
20	30.2 (18.1,39.2)	69.8 (60.8,81.9)	99.0 (93.9,99.0)	1.0 (1.0,6.1)	99.9 (99.7,100)	0.1 (0,0.3)	100 (99.9,100)	0 (0,0.1)	100 (100,100)	0 (0,0)

Note: The trend and cycle denote the relative contribution of trend and cycle innovations. The brackets shows 68% credible intervals computed from 25000 draws. The analysis is performed on the data ranging from 1960 (Q1) - 2022 (Q3).

Finally, we also used historical decomposition to display the contribution of individual shocks. We focus only on the cyclical components as obtaining economically meaningful structural shocks using Choleski decomposition regarding trend components is practically impossible.¹⁸ Regarding cyclical components, we use Choleski decomposition with the following standard ordering $(y_t, c_t, \pi_t, \pi_t^e, i_t)$. In other words, output is ordered first and interest rate is ordered last and does not react contemporaneously to other shocks.

The results are displayed in Figure 3. Black lines denote median estimates. Red bars are output shocks. Yellow bars denote consumption shocks. Green bars are inflation shocks. Blue bars are inflation expectation shocks and purple bars denote interest rate innovations. Our results show that the development of the cyclical components of output, consumption, and interest rates is primarily influenced by output innovations. The inflation gap can be decomposed into the contribution from output shocks and inflation shocks as contributions of other series are negligible. In a similar way, the behavior of the inflation expectations gap is explained by output shocks and its own innovations. However, given that inflation and inflation expectations load upon the inflation trend, which closely corresponds with the inflation expectations series, both gap series are relatively volatile and could contain a large portion of high-frequency noise.¹⁹

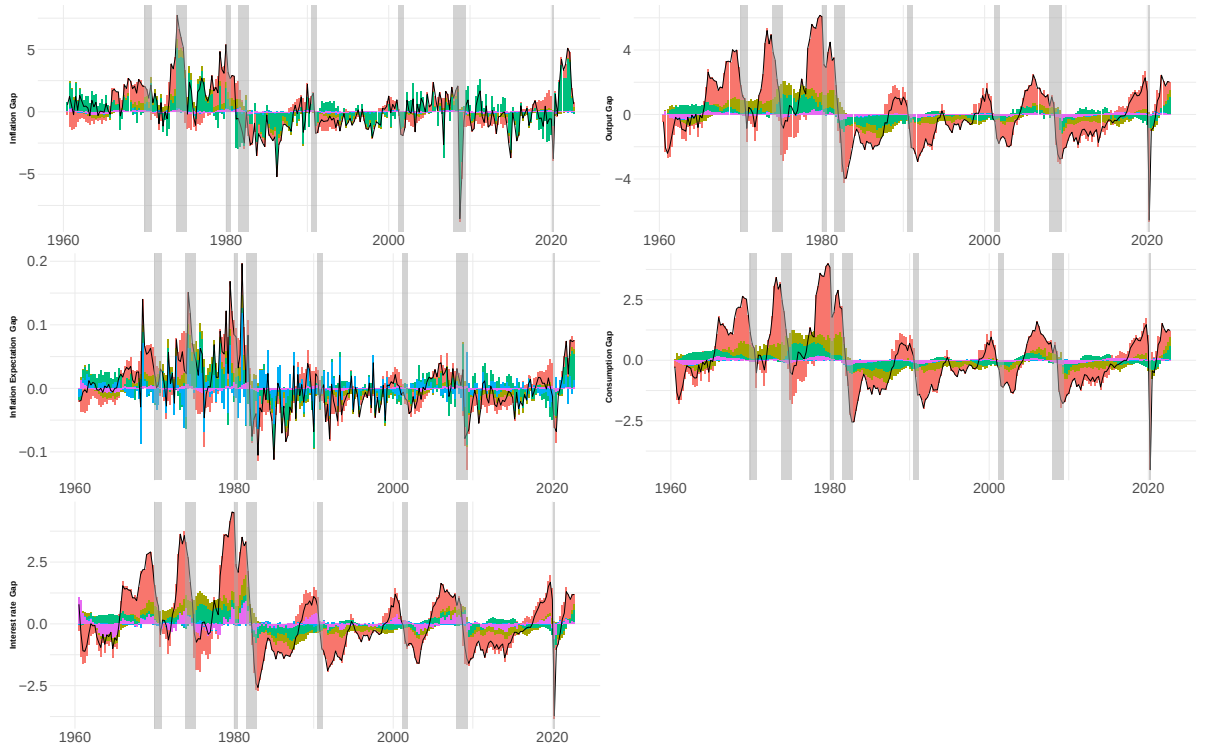
These findings suggest that output innovations are the dominant factor causing business cycle fluctuations. In a similar fashion, Andrieu *et al.* (2017) show that regarding business

¹⁸For alternative approach, see Maffei-Faccioli (2021).

¹⁹Also note that the inflation in annualized quarterly rate, given the fact that central banks mostly focus on year-on-year rate this phase shift may influence the results.

cycle fluctuations, there is mostly positive co-movement of output, interest rate, and inflation. The authors explain the results obtained by the fact that the main driver of business cycle fluctuations are demand shocks. In our case, the result is not as clear. However, our results suggest that monetary policy is highly predictable as the share of monetary shocks is negligible. We also show that monetary policy does not significantly react to inflationary shocks. This is in line with the finding by Bauer *et al.* (2022), where authors find the inflation coefficient close to zero, which is in contrast with the standard 'Taylor principle'. They offer an explanation that the output gap is a sufficient statistic for inflationary pressures, and variation in the perceived output gap coefficient captures the Fed's policy rule. In a similar way, we believe that our results can be explained in the same fashion. The business cycle fluctuations are dominated by demand (output) shocks, which is sufficient statistics for monetary policy rule as the monetary policy does not respond to supply shock as is documented in Časta (2022).

Figure 3: Historical Decomposition - Cyclical Components



Note: Black line - median. Red bar - Output shocks. Yellow bar - Consumption shocks. Green bar - Inflation shocks. Blue bar - Inflation expectation shocks. Purple bar - Interest rate shocks. The analysis is performed on the data ranging from 1960 (Q1) - 2022 (Q3).

6 Conclusion

In this article, we use common trend VAR to extract trend and cyclical components of output, consumption, inflation, and interest rates. We apriori account for cointegration relations based on economic theory. Specifically, we motivate our model specification by the permanent income hypothesis and consumption Euler equation. The main contribution is the identification and

interpretation of these components and subsequent estimation of the natural/equilibrium rate of interest.

Specifically, we present evidence that the trend growth rate of output and consumption (total factor productivity growth rate) is relatively volatile as it is constantly being changed by new shocks, which is in line with Real Business Cycle (RBC) reasoning. The trend cycle decomposition also allows us to study interactions between trend and cyclical components. Specifically, we demonstrate that most of the variance regarding one-step-ahead prediction errors regarding consumption is due to permanent shocks. This sharply contrasts with output results regarding one-step-ahead prediction errors where contributions of cyclical and permanent shocks are divided approximately equally. In summary, using FEVD we generally show that output and inflation contain statistically and economically important transitory components. The FEVD regarding other variables used in the model is then dominated by permanent shocks. The historical decomposition results further show that the behavior of cyclical components is mostly dominated by output shocks, which also appear to be sufficient statistics for monetary policy rule. The contribution of output innovations is lowest in the case of inflation and inflation expectations. However, inflation and inflation expectations load upon the inflation trend. The cyclical components estimates are relatively volatile and could contain high-frequency noise.

Regarding the equilibrium rate of interest, we show that the equilibrium interest rate is highly volatile due to the volatile trend growth rate of output and consumption (total factor productivity growth rate) as our approach is based on the Euler equation, which suggests a cointegration relation exists between the consumption growth rate and the real interest rate. However, we also document the declining trend growth rate of output and consumption can be considered an important contributor to the declining equilibrium interest rate. Nevertheless, the decrease in the trend growth rate of output and consumption is not strong enough to explain the decline in the equilibrium interest rate in the last two decades alone. The decline in this period is mainly explained by a decrease in the additional long-term trend regarding real interest rate evolution, which serves as a proxy for other factors influencing equilibrium rate development.

References

- ANDRLE, M., J. BRUHA, & M. S. SOLMAZ (2013): *Inflation and output comovement in the euro area: Love at second sight?* International Monetary Fund.
- ANDRLE, M., J. BRUHA, & S. SOLMAZ (2017): “On the sources of business cycles: implications for DSGE models.” *Working Paper Series 2058*, European Central Bank.
- ATTFIELD, C. & J. R. TEMPLE (2010): “Balanced growth and the great ratios: New evidence for the us and uk.” *Journal of Macroeconomics* **32**(4): pp. 937–956.
- BALL, L. & N. G. MANKIW (2023): “Market power in neoclassical growth models.” *The Review of Economic Studies* **90**(2): pp. 572–596.
- BAUER, M. D., C. PFLUEGER, & A. SUNDERAM (2022): “Perceptions about monetary policy.” *Technical report*, National Bureau of Economic Research.
- BAUER, M. D. & G. D. RUDEBUSCH (2020): “Interest rates under falling stars.” *American Economic Review* **110**(5): pp. 1316–1354.
- BENEŠ, J., K. CLINTON, R. GARCIA-SALTOS, M. JOHNSON, D. LAXTON, P. B. MANCHEV, & T. MATHESON (2010): “Estimating potential output with a multivariate filter.” .
- BUNCIC, D. (2021): “On a standard method for measuring the natural rate of interest.” *arXiv preprint arXiv:2103.16452* .
- CAMPBELL, J. Y. & J. H. COCHRANE (1999): “By force of habit: A consumption-based explanation of aggregate stock market behavior.” *Journal of political Economy* **107**(2): pp. 205–251.
- CAMPBELL, J. Y. & N. G. MANKIW (1989): “Consumption, income, and interest rates: Reinterpreting the time series evidence.” *NBER macroeconomics annual* **4**: pp. 185–216.
- CARROLL, C. D. (2022): “Theoretical foundations of buffer stock saving.” .
- CARROLL, C. D., R. E. HALL, & S. P. ZELDES (1992): “The buffer-stock theory of saving: Some macroeconomic evidence.” *Brookings papers on economic activity* **1992**(2): pp. 61–156.
- CARROLL, C. D. & L. H. SUMMERS (1991): “Consumption growth parallels income growth: Some new evidence.” In “National saving and economic performance,” pp. 305–348. University of Chicago Press.
- ČASTA, M. (2022): “Supply shocks, demand shocks and yield curve dynamics.” *Finance Research Letters* **50**: p. 103262.
- CERRA, V., A. FATÁS, & S. C. SAXENA (2023): “Hysteresis and business cycles.” *Journal of Economic Literature* **61**(1): pp. 181–225.
- CESA-BIANCHI, A., R. HARRISON, & R. SAJEDI (2022): “Decomposing the drivers of Global R.” *Bank of England working papers 990*, Bank of England.
- CHRISTIANO, L. J. & T. J. FITZGERALD (2003): “The band pass filter.” *international economic review* **44**(2): pp. 435–465.
- CIESLAK, A. & P. POVALA (2015): “Expected returns in treasury bonds.” *The Review of Financial Studies* **28**(10): pp. 2859–2901.
- COCHRANE, J. H. (1994): “Permanent and transitory components of gnp and stock prices.” *The Quarterly Journal of Economics* **109**(1): pp. 241–265.
- DEATON, A. (1992): *Understanding consumption*. Oxford University Press.
- DEL NEGRO, M., D. GIANNONE, M. P. GIANNONI, & A. TAMBALOTTI (2017): “Safety, liquidity, and the natural rate of interest.” *Brookings Papers on Economic Activity* **2017**(1): pp. 235–316.
- DEL NEGRO, M., D. GIANNONE, M. P. GIANNONI, & A. TAMBALOTTI (2019): “Global trends in interest rates.” *Journal of International Economics* **118**: pp. 248–262.
- ELMINEJAD, A., T. HAVRÁNEK, & Z. HAVRÁNKOVÁ (2022): “People are less risk-averse than economists think.” *Technical report*, IES Working Paper.
- ENGLE, R. F. & C. W. GRANGER (1987): “Cointegration and error correction: representation, estimation, and testing.” *Econometrica: journal of the Econometric Society* pp. 251–276.
- GARRATT, A., D. ROBERTSON, & S. WRIGHT

- (2006): “Permanent vs transitory components and economic fundamentals.” *Journal of Applied Econometrics* **21**(4): pp. 521–542.
- HALL, R. E. (1978): “Stochastic implications of the life cycle-permanent income hypothesis: theory and evidence.” *Journal of political economy* **86**(6): pp. 971–987.
- HAVRANEK, T., R. HORVATH, Z. IRSOVA, & M. RUSNAK (2015): “Cross-country heterogeneity in intertemporal substitution.” *Journal of International Economics* **96**(1): pp. 100–118.
- HODRICK, R. J. & E. C. PRESCOTT (1997): “Post-war us business cycles: an empirical investigation.” *Journal of Money, credit, and Banking* pp. 1–16.
- HOLSTON, K., T. LAUBACH, & J. C. WILLIAMS (2017): “Measuring the natural rate of interest: International trends and determinants.” *Journal of International Economics* **108**: pp. S59–S75.
- KILEY, M. T. (2020a): “The global equilibrium real interest rate: Concepts, estimates, and challenges.” *Annual Review of Financial Economics* **12**: pp. 305–326.
- KILEY, M. T. (2020b): “What Can the Data Tell Us about the Equilibrium Real Interest Rate?” *International Journal of Central Banking* **16**(3): pp. 181–209.
- KIM, C.-j., J. M. PIGER, & R. STARTZ (2007): “The dynamic relationship between permanent and transitory components of us business cycles.” *Journal of Money, Credit and Banking* **39**(1): pp. 187–204.
- KING, R. G., C. I. PLOSSER, & S. T. REBELO (1988): “Production, growth and business cycles: I. the basic neoclassical model.” *Journal of monetary Economics* **21**(2-3): pp. 195–232.
- KING, R. G., C. I. PLOSSER, & S. T. REBELO (2002): “Production, growth and business cycles: Technical appendix.” *Computational Economics* **20**: pp. 87–116.
- KING, R. G., C. I. PLOSSER, J. H. STOCK, & M. W. WATSON (1991): “Stochastic trends and economic fluctuations.” *The American Economic Review* pp. 819–840.
- KOZICKI, S. & P. A. TINSLEY (2001): “Shifting endpoints in the term structure of interest rates.” *Journal of monetary Economics* **47**(3): pp. 613–652.
- LAUBACH, T. & J. C. WILLIAMS (2003): “Measuring the natural rate of interest.” *Review of Economics and Statistics* **85**(4): pp. 1063–1070.
- MAFFEI-FACCIOLI, N. (2021): “Identifying the sources of the slowdown in growth: Demand vs. supply.” .
- MIAN, A. R., L. STRAUB, & A. SUFI (2021): “What explains the decline in r^* ? rising income inequality versus demographic shifts.” *Rising Income Inequality Versus Demographic Shifts (September 22, 2021)*. University of Chicago, Becker Friedman Institute for Economics Working Paper (2021-104).
- MORLEY, J., D. RODRÍGUEZ-PALENZUELA, Y. SUN, & B. WONG (2023): “Estimating the euro area output gap using multivariate information and addressing the covid-19 pandemic.” *European Economic Review* **153**: p. 104385.
- MORLEY, J. & B. WONG (2020): “Estimating and accounting for the output gap with large bayesian vector autoregressions.” *Journal of Applied Econometrics* **35**(1): pp. 1–18.
- MORLEY, J. C. (2007): “The slow adjustment of aggregate consumption to permanent income.” *Journal of Money, Credit and Banking* **39**(2-3): pp. 615–638.
- MURASAWA, Y. (2015): “The multivariate beveridge-nelson decomposition with $i(1)$ and $i(2)$ series.” *Economics Letters* **137**: pp. 157–162.
- RACHEL, L. & T. D. SMITH (2017): “Are Low Real Interest Rates Here to Stay?” *International Journal of Central Banking* **13**(3): pp. 1–42.
- SMETS, F. & R. WOUTERS (2007): “Shocks and frictions in us business cycles: A bayesian dsge approach.” *American economic review* **97**(3): pp. 586–606.
- SOMMER, M. (2007): “Habit formation and aggregate consumption dynamics.” *The BE journal of macroeconomics* **7**(1).
- STOCK, J. H. & M. W. WATSON (1988): “Testing for common trends.” *Journal of the American statistical Association* **83**(404): pp. 1097–1107.

- TÓTH, M. (2021): “A multivariate unobserved components model to estimate potential output in the euro area: a production function based approach.” .
- VILLANI, M. (2009): “Steady-state priors for vector autoregressions.” *Journal of Applied Econometrics* **24**(4): pp. 630–650.
- WARNE, A. (1993): *A common trends model: identification, estimation and inference*. IIES.
- WHELAN, K. (2003): “A two-sector approach to modeling us nipa data.” *Journal of Money, Credit and Banking* pp. 627–656.